



Molecular dynamics studies on interfacial interactions between imidazolium-based ionic liquids and carbon nanotubes

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Abstract

We have investigated the basic mechanism of carbon nanotube (CNT) interactions with various room-temperature ionic liquids (RTILs) using molecular dynamics (MD) simulations. To understand the effects of the cation molecular geometry on the properties of the interface structure in the RTIL systems, we have studied a set of three RTILs with the same $[\text{BF}_4]^-$ (tetrafluoroborate) anion but with different cations, namely, $[\text{EMIM}]^+$ (1-ethyl-3-methylimidazolium), $[\text{BMIM}]^+$ (1-butyl-3-methylimidazolium), $[\text{HMIM}]^+$ (1-hexyl-3-methylimidazolium), and $[\text{OMIM}]^+$ (1-octyl-3-methylimidazolium) ions. The simulation results showed that the imidazolium cations exhibit two distinct orientations (perpendicular and parallel to the CNTs surface) at the interface irrespective of the alkyl chain length of the cations. The average number of hydrogen bonds per cations inside the CNT was found to be higher for $[\text{OMIM}][\text{BF}_4]$ (1.01), which suggests that $[\text{OMIM}]^+$ imidazolium rings to be concentrated at the center of the CNT, which favors hydrogen bond. The reported results show the diffusion coefficients of ions in confinement are much lower in comparison to the bulk region. The interaction energy between $[\text{OMIM}][\text{BF}_4]$ ($-8.75 \text{ kcal.mol}^{-1}.\text{ion}^{-1}$) and CNT was found to be higher as compared to other ILs. The cations paralleling the CNT surface are thermodynamically significantly more stable because of the substantial interfacial π - π stacking interactions, as shown by a comparison with the calculated interaction energies between cations and the CNTs. Our simulation results provide a molecular-level understanding of the stabilization and dispersion of CNT bundles in ILs.

Keywords Ionic liquids · Carbon nanotubes · Interface · Molecular dynamics · Interaction energy

Introduction

Carbon nanotubes (CNTs) have gained a lot of attention in recent decades due to their remarkable properties and potential application in energy storage, optical, electrical, sensors, and other fields [1–3]. Due to their strong van der Waals interactions, carbon nanotubes have limited dispersity in organic or water-based solvents, which severely restricts their use in practical preparation and even applications [4]. Imidazolium-based ILs have been shown through numerous experimental studies to be capable of substituting organic solvents in the production of different CNTs with stable dispersions [5–7]. These unique interactions between ILs and CNTs provide a simple method of obtaining

high-performance CNTs without changing their molecular structure. Although research on the interactions between ionic liquids and carbon nanostructures is still in its infancy, it has primarily concentrated on two areas: (i) investigating the properties of mixtures containing graphene or carbon nanotubes, and (ii) investigating the characteristics of ionic liquids that are contained within carbon nanotubes. Studies have demonstrated that confined room temperature ionic liquids (IL) exhibit interesting characteristics [8, 9].

Recent experimental findings about the mixing properties of IL and CNTs have demonstrated that ILs may both disseminate large amounts of CNTs and untangle them into extremely fine bundles [10]. It is essential to comprehend this dispersion mechanism in order to advance CNT-based technology. Molecular dynamics (MD) simulations can considerably contribute to a better understanding of dispersion mechanisms by precisely simulating the IL-CNT interfacial interactions [11, 12]. The dispersion of CNTs in ILs was examined by Mohammadi and coauthors [13]. They discovered that non-bundled CNTs can be dispersed by ILs. When CNTs are not

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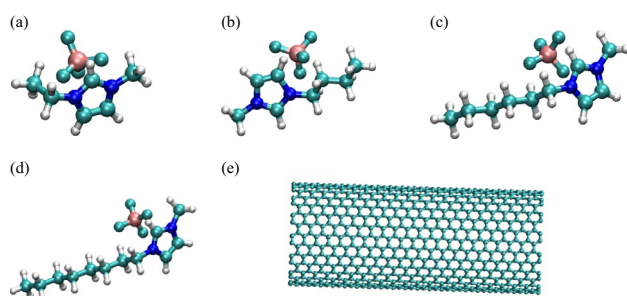


Fig. 1 Molecular structures of [EMIM][BF₄] (a), [BMIM][BF₄] (b), [HMIM][BF₄] (c), [OMIM][BF₄] (d), and (15,15) CNT (length=50 Å, diameter=20.5 Å) (e)

bundled, their solvent-nanotube contact surface is larger than when they are bundled. Larger contact areas indicate that π -stacking interactions between CNT and solvent increased, weakening CNT-CNT interactions. The scenario is the opposite with bundled CNTs. In order to study the solvation structures and dynamical characteristics of imidazolium-based ionic liquids surrounding single-walled CNTs, Yang et al. also carried out a number of MD simulations [14, 15]. Their findings showed that cations' imidazolium rings prefer to lie nearly parallel to the surface of CNTs, which greatly increased the hydrogen bonds between cations and anions at the interface. Despite significant efforts, MD simulations do not directly reveal the nature of the non-covalent interactions between interface imidazolium-based ILs and CNTs. Huo and Liu noted that at the interface of the armchair (6,6) single-walled CNTs, both [BMIM]⁺ cations and [BF₄][−] anions exhibit a highly ordered orientation [16]. Additionally, they noticed that the interfacial cations exhibited two distinct orientation patterns: one with the imidazolium rings of the cations oriented parallel to the surface,

Table 1 Nonbonded parameters for 1-Alkyl-3-methylimidazolium [RMIM] cations and [BF₄] anions

ILs	Atom type	q(e)	σ (Å)	ϵ (kcal/mol)
Cation [RMIM] ⁺	CR	−0.09	3.55	0.070
	NA	0.22	3.25	0.170
	CW	−0.24	3.55	0.070
	CM	−0.35	3.50	0.066
	CA	−0.17	3.50	0.066
	CS	−0.12	3.50	0.066
	CT	−0.24	3.50	0.066
	HR	0.21	2.42	0.030
	HW	0.27	2.42	0.030
	HM	0.18	2.50	0.030
	HA	0.18	2.50	0.030
	HS	0.06	2.50	0.030
	HT	0.08	2.50	0.030
Anion [BF ₄] [−]	B	0.82	3.58	0.095
	F	−0.45	3.11	0.060

Table 2 Average number of cations and anions, N , inside the CNT at 300 K

Systems	N_{Cation}	N_{Anion}
[EMIM][BF ₄]	44	44
[BMIM][BF ₄]	32	32
[HMIM][BF ₄]	24	27
[OMIM][BF ₄]	18	22

and another with the rings oriented perpendicular to the surface. In order to investigate the adsorption behavior of N-ethyl-N-(furan-2-ylmethyl)ethaniminium dihydrogen phosphate ILs at the interface of graphene and CNTs, García and co-workers used MD simulations together with density functional theory (DFT) [17]. They observed that the adsorption of ILs on graphene or CNTs is driven by van der Waals interactions. Atilhan and Aparicio [18] found that π - π stacking characterizes the interaction between the cation and graphene and it is nearly three times stronger than the anion-graphene interaction. New interaction models for the interface between imidazolium-based ILs and SWCNTs were studied by França et al. [19] They used the results of electronic structure calculations to fit n-m interaction potential functions (n varying between 9 and 10 and m varying between 5 and 8) in order to better describe interactions like those between ions and highly polarizable carbon nanotubes, which they believed could not be adequately described by simple Lennard–Jones potentials ($n=6$, $m=12$) with parameters derived from mixing rules. Nevertheless, one of their objectives was the calculation of thermal conductivity of the composite systems using non-equilibrium MD simulations. Peng et al. [20] studied the relationship between interfacial interactions and nanostructures of [BMIM][BF₄] IL electrolyte surrounding single-walled carbon nanotube electrodes with different charge densities and nanotube diameters. They have mentioned that the imidazolium rings of cations around negative or positive electrodes with lower charge density have more ordered orientations than those with higher charge density.

In this work, MD simulations were used to study combinations of imidazolium-based ILs with CNTs. One of the main objectives is to understand the impacts of imidazolium cation alkyl length with common anion on the relevant interfacial structure and properties that have been investigated. The simulations of [EMIM][BF₄], [BMIM][BF₄], [HMIM]

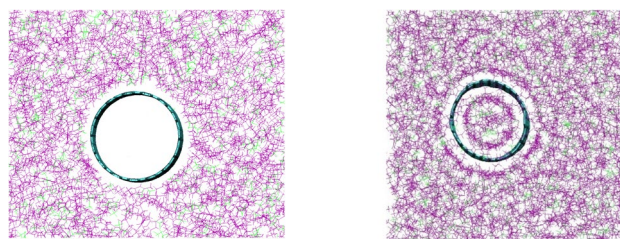


Fig. 2 The initial and final snapshots of [OMIM][BF₄]-CNT system

[BF₄], and [OMIM][BF₄] mixtures with a (15,15) CNT were performed. The interfacial layering structure of the four ILs was studied, with a particular emphasis on the influence of the cation's alkyl groups' "tails" size on the nanostructure of the external and internal layers. A detailed discussion is held over the orientation of the anions and cations on the interface between CNT and IL. The composition of the outside and inside surface of the CNT as well as the interaction energy between the CNT and IL are important for understanding the stabilization mechanism of CNTs in ILs.

Computational approach

We used 1-ethyl-3-methylimidazolium tetrafluoroborate [EMIM][BF₄], 1-n-butyl-3-methylimidazolium tetrafluoroborate [BMIM][BF₄], 1-hexyl-3-methylimidazolium tetrafluoroborate [HMIM][BF₄], and 1-octyl-3-methylimidazolium tetrafluoroborate [OMIM][BF₄] in this study. Figure 1 shows the molecular structures of the ILs. The length of the alkyl "tail" is the primary distinction between the ILs. A (15,15) armchair nanotube with 1260 carbon atoms, with a length of $L = 50$ Å, and a diameter of $D = 20.5$ Å was used in a mixture of CNTs with ILs. A depiction of this nanotube is shown in Fig. 1e. We conducted MD simulations of CNT dissolved in several RTILs in order to reveal fundamental molecular mechanisms of ion interactions with CNT surfaces. In our simulations, we used the completely atomistic OPLS-AA force field parameters [21, 22]. The force field parameters for ILs were obtained from Lopes et al. [23, 24]. The partial atomic charges were calculated by the CHELPG method [25], whereas the nonbonded interactions were

determined using a 12 Å atom-based cutoff. The nonbonded parameters and the partial atomic charges for ILs are mentioned in Table 1. To assess the electrostatic interactions, we employed the Particle–Mesh Ewald method [26].

We used NAMD software [27] to run the MD simulations. The VMD software [28] was used to produce the CNT atoms' coordinates. The carbon atom of the CNT was considered an uncharged particle. The Packmol software [29] was used to distribute 600 IL pairs randomly inside the simulation box with $65 \times 65 \times 65$ Å³ (none were placed inside the nanotubes). A single-walled carbon nanotube (CNT) was placed in the center of a simulation box as shown in Fig. 1e. The simulation boxes were then equilibrated within the framework of an isothermal-isobaric (NPT) ensemble with periodic boundary conditions. The Langevin piston and thermostat were used to control the pressure and temperature. The equations of motion were integrated with a time step of 1.0 fs using the Verlet algorithm. The system was subjected to five cycles of annealing at NVT in the temperature range of 300–650 K for a total duration of 100 ps. After the cyclic annealing the system was equilibrated at the NPT ensemble for 5 ns at 300 K and 1 atm. Once it was confirmed that the ILs had filled the nanotubes for at least 1 ns, a second equilibration run was carried out for 5 ns at 300 K and 1 atm. The densities of the simulation boxes were calculated after the last equilibration run. The densities of [EMIM][BF₄], [BMIM][BF₄], [HMIM][BF₄], and [OMIM][BF₄] systems in the presence of CNT are 1.291, 1.204, 1.148, and 1.125 g/cm³, respectively, which are in close agreement with experimental works [30–32]. The production runs for the mixtures lasted 20 ns and the data on trajectory was collected at intervals of 1 ps.

Fig. 3 Number density profiles along the radial direction of ions confined inside CNT for [EMIM][BF₄] (a), [BMIM][BF₄] (b), [HMIM][BF₄] (c), and [OMIM][BF₄] (d) systems

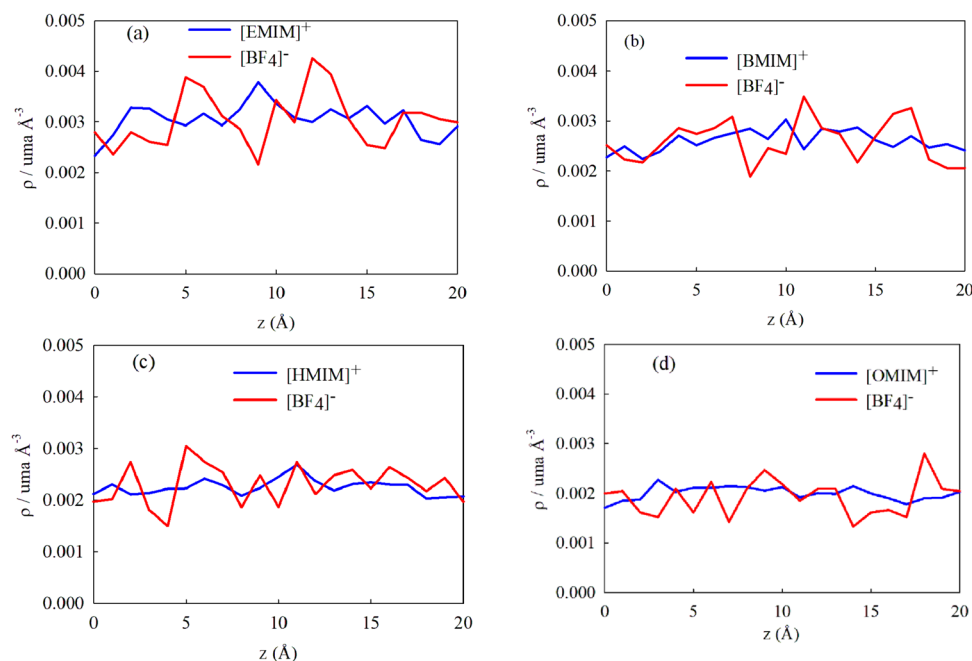
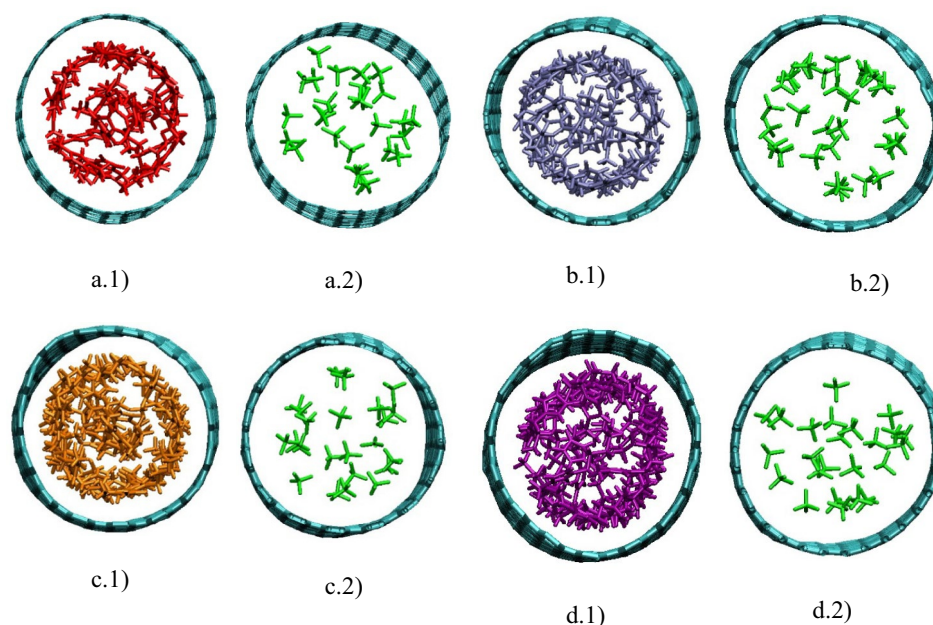


Fig. 4 The spatial disposition of [EMIM]⁺ (red) and [BF₄][−] (green) (a), [BMIM]⁺ (ice blue) and [BF₄][−] (green) (b), [HMIM]⁺ (orange) and [BF₄][−] (green) (c), and [OMIM]⁺ (purple) and [BF₄][−] (green) (d) inside of the CNT



Results and discussion

In this section, we will investigate the changes in ionic liquid structuring caused by confinement within CNTs, as well as the behavior of ILs on the outer curved surface of CNTs.

Understanding the transportation of ILs through CNT

We have investigated the changes in ionic liquid structure caused by confinement within CNTs. Table 2 shows the average number of ions within the CNT. The number of

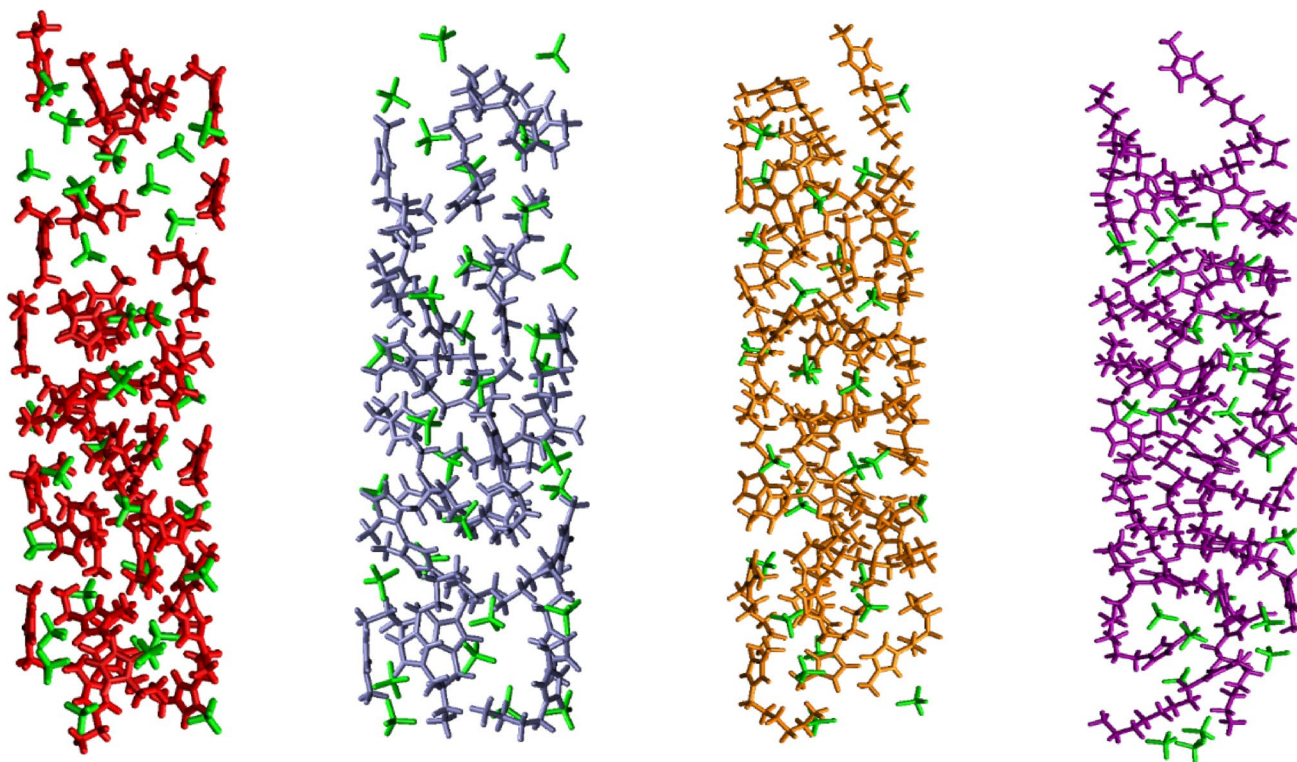


Fig. 5 Zoom of the spatial disposition of [EMIM][BF₄] (a), [BMIM][BF₄] (b), [HMIM][BF₄] (c), and [OMIM][BF₄] (d). Molecules are colored according to the color scheme used in Fig. 4

ions within the CNT is the same, except that the concentrations of $[\text{HMIM}]^+$ and $[\text{OMIM}]^+$ cations are somewhat lower in CNTs, resulting in virtually electrically neutral distributions. The slightly lower cation number in comparison with anion values for $[\text{HMIM}][\text{BF}_4]$ and $[\text{OMIM}][\text{BF}_4]$ may be explained by considering the large size of the cations. The snapshots for $[\text{OMIM}][\text{BF}_4]$ based system from simulations at the starting and after 20 ns are shown in Fig. 2. The transport mechanism of studied ILs through CNT was analyzed. Density along the CNT radial direction is reported in Fig. 3. For $[\text{EMIM}][\text{BF}_4]$ systems, we obtained the higher density for cations and anions in the center of the nanotube, whereas one and two sharp peaks are found for $[\text{EMIM}]^+$

and $[\text{BF}_4]^-$ ranges from 5 to 13 Å. The densities of $[\text{EMIM}]^+$ and $[\text{BF}_4]^-$ ions are found to be lower near the surface of the nanotube. For the $[\text{BMIM}][\text{BF}_4]$ density profile, we found one sharp peak for $[\text{BMIM}]^+$ at 10 Å (center of the nanotube) and three peaks for $[\text{BF}_4]^-$ at 7 Å, 10 Å, and 17 Å. The density of $[\text{BMIM}]^+$ near the surface of the CNT was found to be higher than that of the center of the CNT. On the other side, no sharp peaks were present for $[\text{HMIM}]^+$ and $[\text{OMIM}]^+$ density profiles in the center of the CNT. The densities of the cations ($[\text{HMIM}]^+$ and $[\text{OMIM}]^+$) were found to be higher near the surface of the CNT. The distributions of cations and anions are also confirmed from the snapshots. Figure 4 reports that ion deposition inside the

Fig. 6 MSD of $[\text{EMIM}]^+$ and $[\text{BF}_4]^-$ at bulk IL phase (a), $[\text{EMIM}]^+$ and $[\text{BF}_4]^-$ inside the CNT (b), $[\text{BMIM}]^+$ and $[\text{BF}_4]^-$ at bulk IL phase (c), $[\text{BMIM}]^+$ and $[\text{BF}_4]^-$ inside the CNT (d), $[\text{HMIM}]^+$ and $[\text{BF}_4]^-$ at bulk IL phase (e), $[\text{HMIM}]^+$ and $[\text{BF}_4]^-$ inside the CNT (f), $[\text{OMIM}]^+$ and $[\text{BF}_4]^-$ at bulk IL phase (g), and $[\text{OMIM}]^+$ and $[\text{BF}_4]^-$ inside the CNT (h)

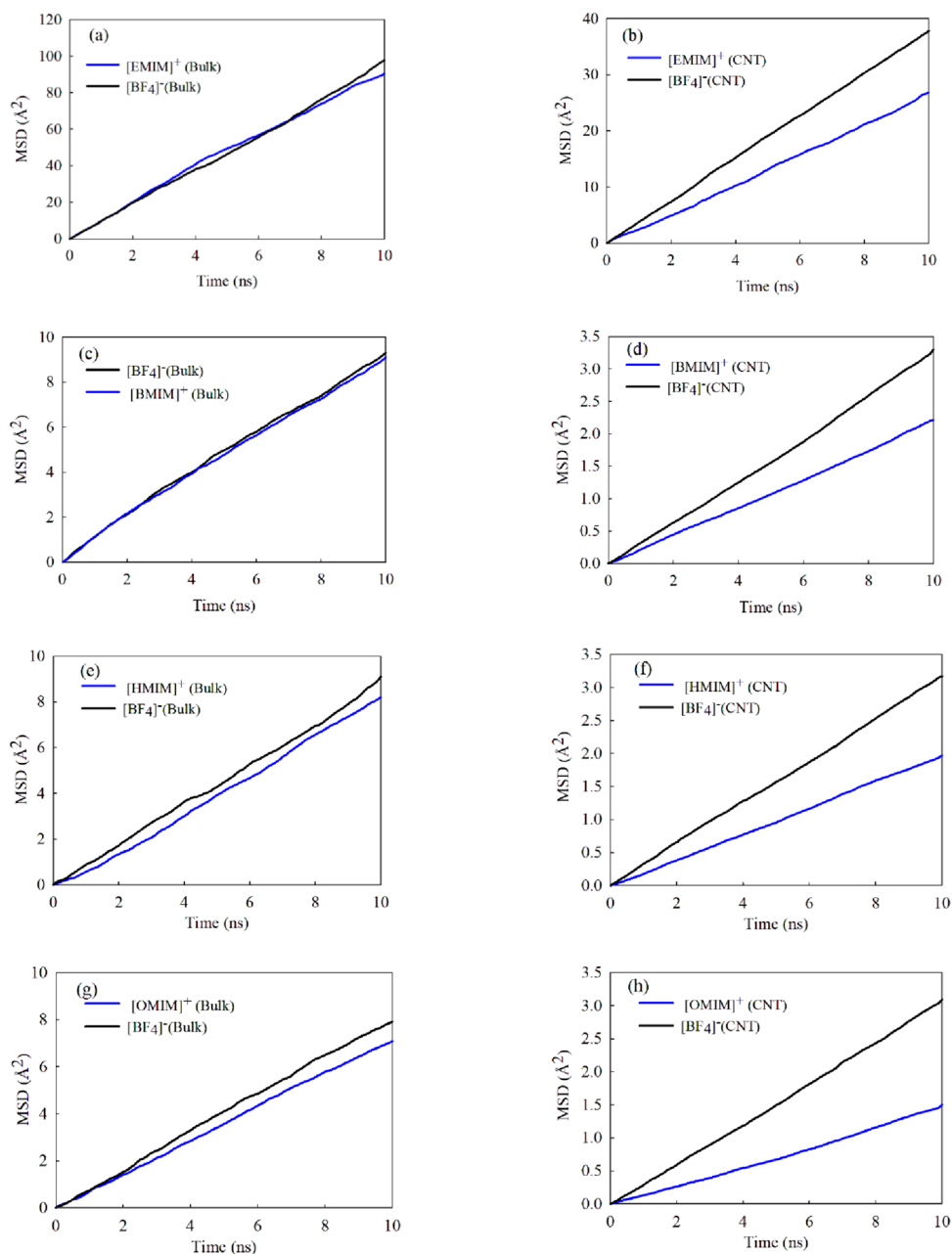


Table 3 Self-diffusion coefficients of anions and cations in bulk phase or inside the CNT

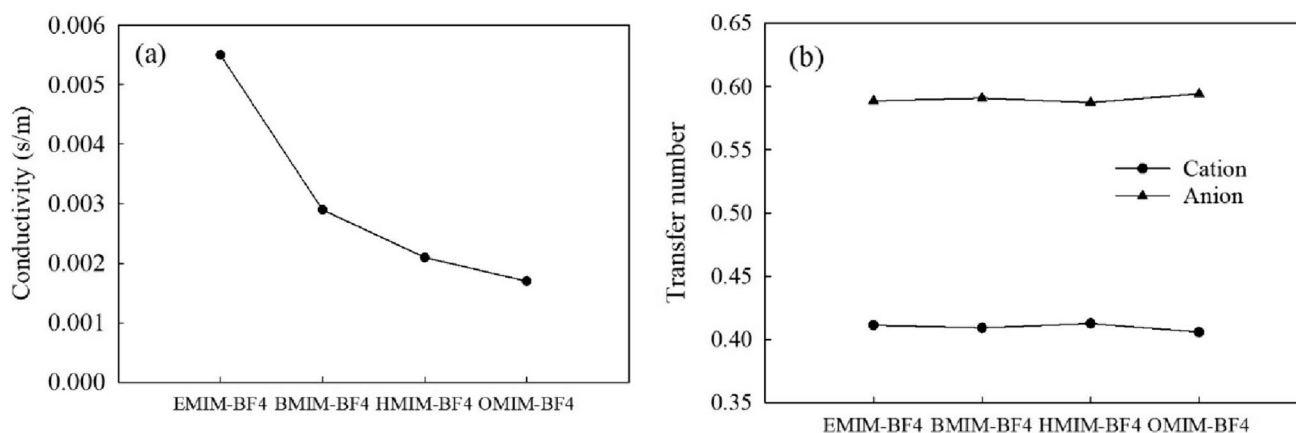
Systems	Species	Phase	Diffusion coefficient $\times 10^{-10}$ (m ² /s)
[EMIM][BF ₄]	[EMIM] ⁺	Bulk	1.50×10^{-10}
	[BF ₄] ⁻	Bulk	1.59×10^{-10}
	[EMIM] ⁺	CNT	0.44×10^{-10}
	[BF ₄] ⁻	CNT	0.63×10^{-10}
[BMIM][BF ₄]	[BMIM] ⁺	Bulk	1.47×10^{-12}
	[BF ₄] ⁻	Bulk	1.51×10^{-12}
	[BMIM] ⁺	CNT	0.36×10^{-12}
	[BF ₄] ⁻	CNT	0.52×10^{-12}
[HMIM][BF ₄]	[HMIM] ⁺	Bulk	1.41×10^{-12}
	[BF ₄] ⁻	Bulk	1.46×10^{-12}
	[HMIM] ⁺	CNT	0.33×10^{-12}
	[BF ₄] ⁻	CNT	0.47×10^{-12}
[OMIM][BF ₄]	[OMIM] ⁺	Bulk	1.20×10^{-12}
	[BF ₄] ⁻	Bulk	1.34×10^{-12}
	[OMIM] ⁺	CNT	0.28×10^{-12}
	[BF ₄] ⁻	CNT	0.41×10^{-12}

CNT is not identical for all ILs. For all IL systems, the anions and cations both occupy the inner surface of the CNT as well as the central area of the nanotube. In the case of the [EMIM][BF₄] system, we found that more cations and anions are placed in the center of the CNT as compared to other ILs. The reported results display two internal shells for [EMIM]⁺ and [BF₄]⁻, which is in agreement with the density behavior reported in Fig. 3a. The distributions of cations such as [HMIM]⁺ and [OMIM]⁺ were lower than that of [EMIM]⁺ and [BMIM]⁺ in the inside of the CNT. Lower densities for [HMIM][BF₄] and [OMIM][BF₄] within CNTs channels compared to [EMIM][BF₄] and [BMIM][BF₄] may

be due to the huge sizes of the relevant ions. Figure 5 provides a better understanding of ionic spatial distribution. It can be seen that [HMIM]⁺ and [OMIM]⁺ imidazolium rings are mostly toward the center of the CNT due to the long alkyl tail traveling parallel to the CNT sidewall. On the other hand, [EMIM]⁺ and [BMIM]⁺ cations have their imidazolium ring preferentially close to the CNT sidewall.

The average number of hydrogen bonds per cation was computed in order to further characterize the ILs inside the tube. The number of hydrogen bonds was obtained considering the probability of hydrogen bonding between hydrogen atoms attached to the imidazolium ring (HW and HR) and F (fluorine) and B (boron) atoms with distance conditions (donor–acceptor distance) of less than 3.5 Å and angle conditions of less than 40° [33]. It was found that in CNT confinement, the average number of hydrogen bonds per cation is lower in [EMIM][BF₄] (0.78) as compared to [BMIM][BF₄] (0.81), [HMIM][BF₄] (0.89), and [OMIM][BF₄] (1.01), which can be related to the fact that [EMIM]⁺ cations have their imidazolium ring (donor atoms) very close to the CNT side wall that can be an impediment to the establishment of hydrogen bond. The presence of a long alkyl tail parallel to the CNT sidewall causes [HMIM]⁺ or [OMIM]⁺ imidazolium rings to be concentrated at the center of the CNT, which favors hydrogen bonds.

The mobility (i.e., rate of mass transfer) of a molecule is defined by the self-diffusion coefficient of the species in the medium, which is theoretically described by the Einstein equation [34, 35]. The self-diffusion coefficients for cations and anions inside the CNT have been measured using the mean square displacement profiles [34, 36] as shown in Fig. 6. The diffusivities of pure ILs are 1.50×10^{-10} m²/s and 1.59×10^{-10} m²/s for the cation and anion, respectively, of [EMIM][BF₄]; 1.47×10^{-12} m²/s and 1.51×10^{-12} m²/s for the cation and anion, respectively, of [BMIM][BF₄]; 1.41×10^{-12} m²/s and 1.46×10^{-12} m²/s for the cation and anion, respectively, of [HMIM][BF₄]; and 1.20×10^{-12} m²/s and 1.34×10^{-12} m²/s for the cation and

**Fig. 7** The ion conductivity (a) and ion transference numbers (b) for the confined ILs

anion, respectively, of [OMIM][BF₄] (Table 3). The diffusion coefficients of IL in the bulk phase follow the order [EMIM][BF₄] > [BMIM][BF₄] > [HMIM][BF₄] > [OMIM][BF₄] which are in close agreement with experimental works [37, 38]. The diffusion coefficients of cations and anions inside the CNT is much lower than that of IL species in the bulk phase. The cations and anions for [EMIM][BF₄] IL move faster ($0.44 \times 10^{-10} \text{ m}^2/\text{s}$ for [EMIM]⁺ and $0.63 \times 10^{-10} \text{ m}^2/\text{s}$ for [BF₄]⁻) than that of other IL species inside the CNT.

We have also calculated the electrical conductivity of the confined ILs using the Nernst–Einstein relation and presented in Fig. 7a.

$$\sigma = \frac{e^2}{K_b T V'} (N_- Z_-^2 D_- + N_+ Z_+^2 D_+) \quad (1)$$

In this equation, N_- and N_+ are the number of anions and cations, e is the charge of the electron, T is the temperature, K_b is the Boltzmann constant, V' is the inner volume of CNT, Z_- and Z_+ are the formal charges on the anion and cation, and D_- and D_+ are the self-diffusion coefficients of anion and cation, respectively. The transference number for cations and anions has been calculated using the following equation and presented in Fig. 7b:

$$t_+ = \frac{D_+}{D_+ + D_-} \quad (2)$$

$$t_- = \frac{D_-}{D_+ + D_-} \quad (3)$$

The ion conductivity value was found to be higher (0.005 S.m^{-1}) for [EMIM][BF₄] IL as compared to other ILs. The increase in the ion conductivity of confined [EMIM][BF₄] IL is due to the increase in its self-diffusion. This result is also due to the decrease in the average number of hydrogen bonds for the confined [EMIM][BF₄]. Figure 7b represents the transference number for the confined anions and cations of ILs. The transference number of anions was higher than that of cations which is also in agreement with the self-diffusion of anions and cations inside the CNT.

Structural properties of the simulated systems

To understand the arrangement of IL molecules around CNT, we calculated the atomic radial distribution function (RDF) for the interactions of CNT with IL molecules. Figure 8a–d shows

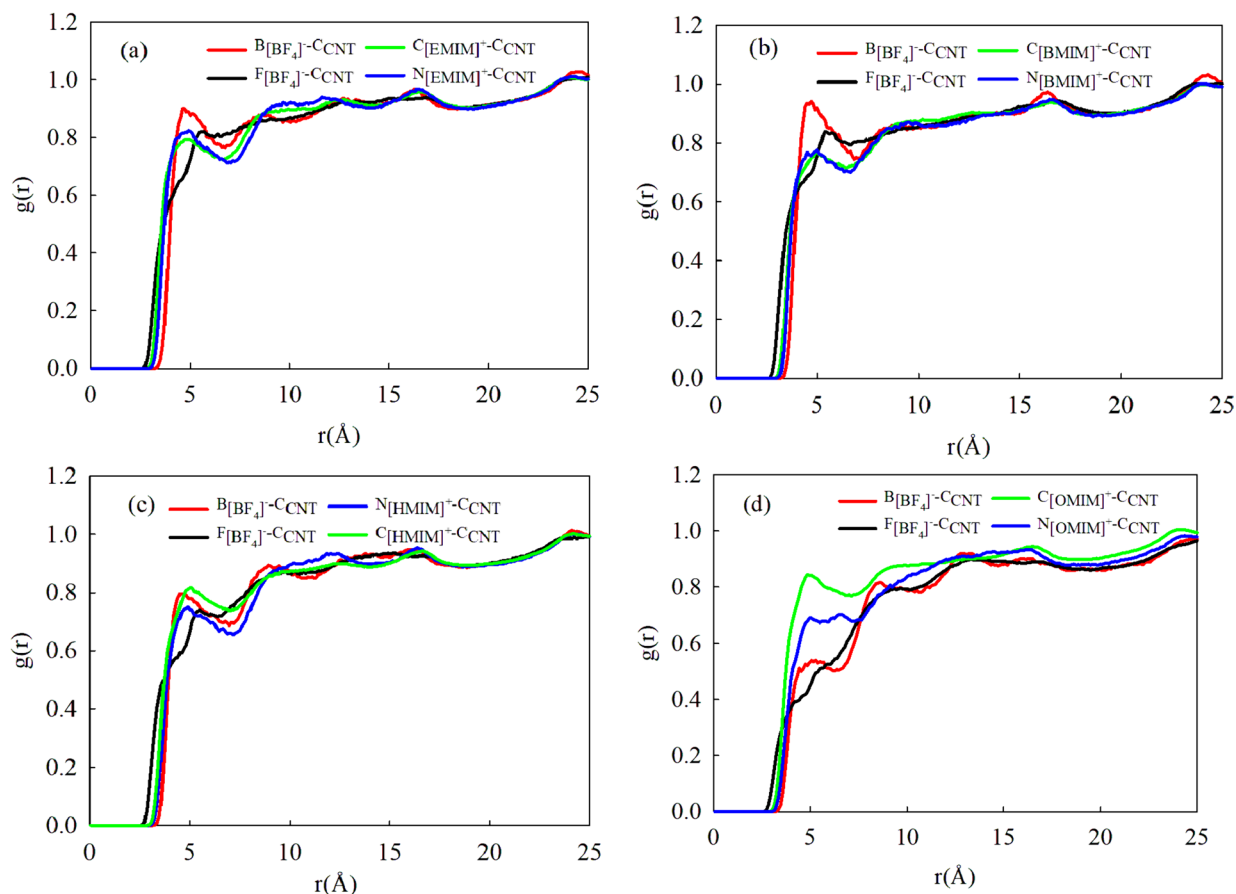


Fig. 8 Radial distribution functions of [EMIM][BF₄] (a), [BMIM][BF₄] (b), [HMIM][BF₄] (c), and [OMIM][BF₄] (d) with CNT

the atomic RDFs for [EMIM][BF₄], [BMIM][BF₄], [HMIM][BF₄], and [OMIM][BF₄] systems at 300 K. We observed a closer interaction of the nitrogen atom of imidazolium ring of [EMIM]⁺ with the carbon atom of CNT (C_{CNT}), which has an RDF peak position at 4.75 Å when comparing the position of the nitrogen atom for other cations ([BMIM]⁺—4.79 Å, [HMIM]⁺—4.83 Å and [OMIM]⁺—5.02 Å). On the other hand, the RDF between the carbon atom of the imidazolium ring of [EMIM]⁺ (C_{[EMIM]⁺}) and C_{CNT} shows the initial peak at 4.91 Å. The RDF peaks for C_{[BMIM]⁺}, C_{[HMIM]⁺}, and C_{[OMIM]⁺} with C_{CNT} are located at 4.90 Å, 4.96 Å, and 4.98 Å, respectively. The π – π stacking interaction peak for all ILs was found to be near about 4.9 Å. This result is also confirmed by the calculating of combined distribution functions (CDFs) between imidazolium rings of interfacial cations and the CNT surface is shown in Fig. 9. θ is the angle formed between two normal vectors, and r is the distance between the center of mass of the imidazolium rings of the cations and the six-membered ring of the CNT. As illustrated in Fig. 9, the π – π stacking probability

distribution is represented by colors ranging from blue to red. Figure 9 shows that there is a noticeable distribution in the region of $r < 5.0$ Å and $\theta > 140^\circ$ for all systems.

We can also understand the structural analysis of cations and anions of IL near the CNT surface by calculating the vertical distances. The vertical distances between the CNT surface and imidazole ring of cations such as [EMIM]⁺, [BMIM]⁺, [HMIM]⁺, and [OMIM]⁺ are 3.11, 3.13, 3.15, and 3.16 Å, respectively, which are all close to the value of isolated cation paralleling to CNT surface (3.10 Å) [39]. Figure 10 shows the orientations of the cations and anions near the surface of CNT. The [EMIM]⁺ and [BMIM]⁺ cations display two different types of orientations such as parallel and perpendicular to the CNTs surface. The large alkyl chain length of cations ([HMIM]⁺ and [OMIM]⁺) are paralleling to the CNT surface. A few numbers of [EMIM]⁺ and [BMIM]⁺ cations display perpendicular orientations due to the substantial CH- π interactions. The positions of [BF₄][−] anion have minimal impact on the orientations of the

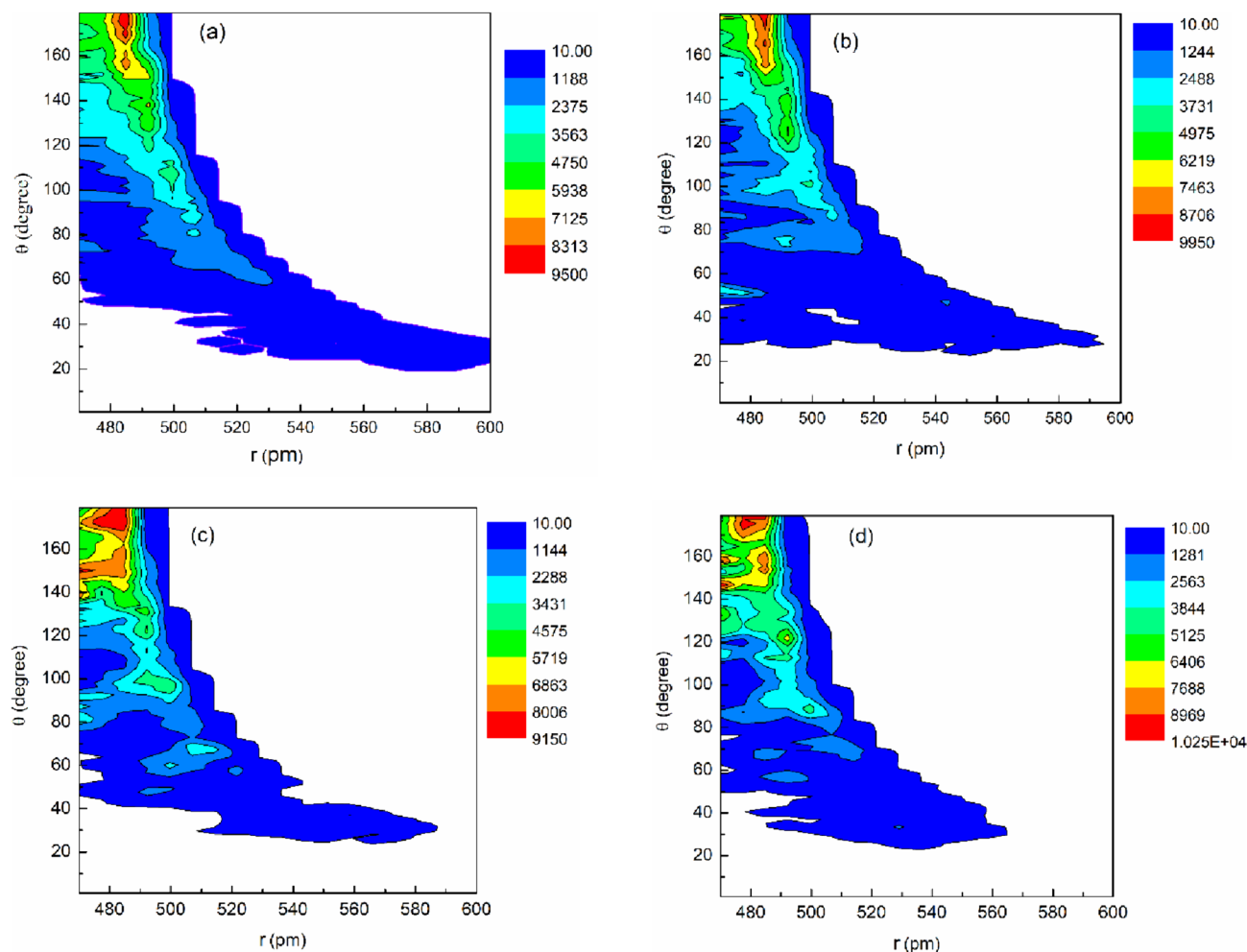


Fig. 9 Combined distribution functions between the imidazolium ring and six-membered ring of CNT in the systems of [EMIM][BF₄] (a), [BMIM][BF₄] (b), [HMIM][BF₄] (c), and [OMIM][BF₄] (d)

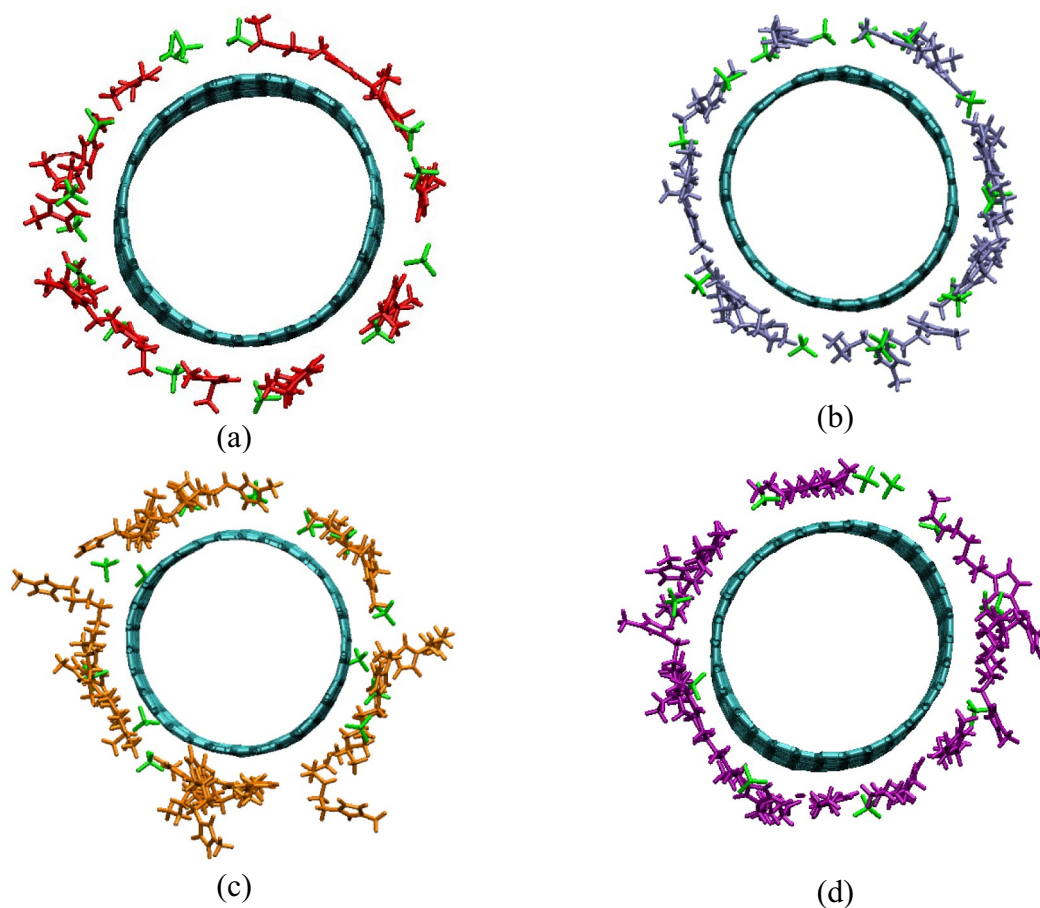


Fig. 10 Snapshots of the orientations of [EMIM][BF₄] (a), [BMIM][BF₄] (b), [HMIM][BF₄] (c), and [OMIM][BF₄] (d) near the interface of CNT

interfacial cations, which are paralleling to the surface of CNT. The main reason for the parallel orientations of the cations to the CNT surface is that it is thermodynamically more stable orientations due to the substantial interfacial π - π stacking interactions. On the other hand, the [BF₄]⁻ anions are discovered to attach to the surface of the CNTs with four fluorine atoms, indicating a highly ordered orientation.

To understand the solvation effect on the surface of CNT, we have determined the interaction energy between CNT and ILs. Figure 11 shows the interaction energy of different systems, obtained by $E_{\text{inter}} = E_{\text{nanotube+IL}} - (E_{\text{nanotube}} + E_{\text{IL}})$; where $E_{\text{nanotube+IL}}$ represents the total energy of the CNT and IL mixed systems, and E_{IL} and E_{nanotube} are the total energies for the pure systems. To determine the interaction energy per ion, the interaction energy was divided by the sum of ions contained inside the nanotubes with those in the shell region. The interaction energy was found to be higher for [OMIM][BF₄] as compared to other ILs which is probably due to the better interaction between the longer alkyl tails of [OMIM]⁺ and the nanotube sidewalls. This result is in good agreement with the self-diffusion

coefficients of the [OMIM]⁺ and [BF₄]⁻ ions inside the CNT. The higher interaction energy and lower diffusivity of [OMIM][BF₄] inside the CNT also suggest better

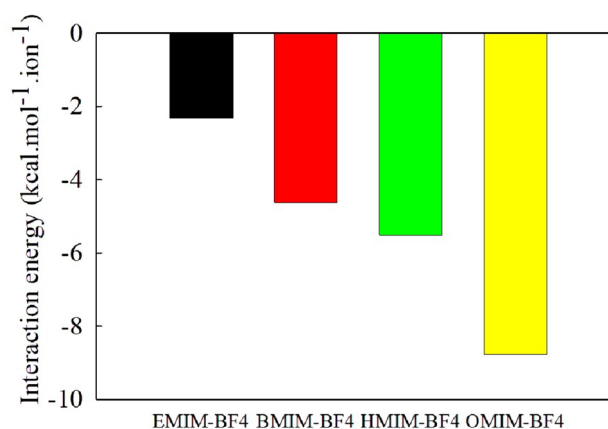


Fig. 11 Interaction energy (per ion) between [EMIM][BF₄] and the CNT, [BMIM][BF₄] and the CNT, [HMIM][BF₄] and the CNT and [OMIM][BF₄] and the CNT

adsorption. We discovered that the orientations of interfacial cations had a significant effect on their interaction energy with the CNTs. In comparison, the interaction energy values of cations paralleling the CNT surface are slightly higher than those perpendicular to the CNT surface, showing that cations paralleling the CNT surface are substantially more thermodynamically stable due to the considerable interfacial π - π stacking interactions.

Conclusions

In this work, the nanoscopic structure of imidazolium-based ionic liquids inside and outside of CNT was investigated using MD simulations. The simulations of selected ILs, [EMIM][BF₄], [BMIM][BF₄], [HMIM][BF₄], and [OMIM][BF₄] were validated by comparing the densities at 300 K and 1 atm and self-diffusion coefficients with data previously published. Then, ILs were separately mixed with CNTs, and their nanostructure was studied using (i) number density profiles along the radial direction of ions confined inside CNT (ii) self-diffusion coefficients, (iii) ion conductivity, (iv) ion transference numbers for the confined ILs, (v) average number of hydrogen bonds, (vi) radial distribution functions, (vii) combined distribution functions, and (viii) interaction energies. The following observations are required:

Three regions may be distinguished in mixtures of ILs and carbon nanotube: a shell region containing ILs ions surrounding the tubes' exterior; a restricted region containing ILs ions inside the tubes; and a bulk region located outside the shell region. These areas have various characteristics. The number density of ions inside the tubes is typically lower than that of the bulk phase. The ion mobility in the shell and confinement regions is significantly lower as compared to the bulk region. Additionally, the number of hydrogen bonds decreases in these areas. IL has a tendency to develop organized layers inside the CNT.

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Author contributions Rima Biswas conceived and designed the study. Prateek Banerjee and Rima Biswas performed simulation calculations. Rima Biswas and Kavathekar Soham Sudesh wrote the paper. All authors read and approved the manuscript.

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Declarations

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